

PUJA KALIVARAPU

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Boston

Results-driven Software Engineer graduating Dec 2025 with 3+ years of production experience building scalable backend systems and user-focused applications. Proven expertise in **Python, Golang, React, and AWS** with hands-on experience architecting microservices that handle 1M+ concurrent requests.

EDUCATION

Northeastern University, Boston

Sept 2023 - Dec 2025

Master of Science in Computer Software Engineering

Courses: OOP Design, Web Development and UX, Cloud Computing, Program Structures and Algorithms

PES University, Bengaluru

July 2016 - July 2020

Bachelor of Technology, Electronics and Communication Engineering (Major), Computer Science (Minor)

Courses: Computer Networking, Data Structures and Algorithms, IoT systems

EXPERIENCE

Fidelity Investments, Boston, MA

Full Stack Engineer Co-op

Jan 2025 – July 2025

- Designed and deployed user-focused full-stack applications using **Python, Streamlit, Pandas**, and **Snowflake** to automate manual, Excel macro-driven workflows across risk and finance teams reducing turn around time by 90% and improving reliability.
- Built robust deployment pipelines using **Jenkins** and **Apache Airflow**, integrating unit testing, code linting, and scheduled execution to ensure consistent delivery and performance.
- Containerized applications with **Docker** and orchestrated deployments on **AWS EC2** instances, enabling scalable, reproducible environments for production use.
- Developed an agentic AI-powered tool using **LangChain** and **LangGraph** that scanned internal Git repositories to answer source code questions, auto-generate unit tests and functionality, and push updates to new branches—accelerating codebase comprehension and refactoring.

Hewlett Packard Enterprise R&D, Bengaluru

System/Software Engineer 1 & 2

Jan 2020 – July 2023

- Designed and deployed scalable, event-driven microservices using **GoLang**, and Python **Flask** on **AWS Kubernetes** clusters, capable of handling over 1 million concurrent update requests.
- Collaborated cross-functionally with DevOps teams** to containerize and deploy microservices on AWS Kubernetes instances, implementing deployment strategies, service mesh configurations, and monitoring solutions that ensured 99.9% uptime across production environments.
- Built robust ETL pipelines to transform XML component data into structured formats and load it into **data-intensive systems** including **Elasticsearch** and **Redis Cache**, leveraging Elasticsearch for optimized database querying, full-text search capabilities, and RabbitMQ messaging system for reliable asynchronous processing—enabling fast and scalable search operations across distributed microservices.
- Automated **CI/CD workflows using Jenkins pipelines**, integrating testing, linting, and deployment to enforce **TDD** and maintain zero-regression standards across releases.
- Participated in agile development methodologies using JIRA for sprint planning, ticket management, and Program Increment (PI) planning, with collaborative story pointing sessions using planning poker techniques.

PROJECTS & PUBLICATIONS

Cloud Native Web Application (REST,Packer IaC, Terraform , GCP, Jest, Node.js, MySQL, CI/CD ,Prisma)

- Implemented scalable cloud infrastructure on **Google Cloud Platform (GCP)** using **Terraform (IaC)** and **Packer**.
- Developed a cloud-native **RESTful APIs with Node.js and Prisma** for user management and health checks, while employing event-driven architecture with **Google Pub/Sub** for messaging and notification services.
- Enhanced security with customer-managed encryption keys (CMEK) for data protection.
- Ensured best **DevOps** performance and availability with load balancing, autoscaling, and continuous deployment (CI/CD) through managed instance groups, utilizing **GitHub Actions**, while emphasizing best practices in security and data management.

TECHNICAL SKILLS

Programming Languages: Python, Golang , Java, JavaScript, HTML/CSS, SQL,C++,Shell/Bash

Data : MySQL, MongoDB, Postgres, Elasticsearch

Back-end & Front-end Frameworks : node.js, Redis, Celery, Express, Flask ,Material UI, React, Redux ,Terraform

Cloud Technologies : AWS EC2,Docker, Kubernetes,GCP